

Dear Editors,

Please consider my manuscript, “Mechanistic equivalence and measurement design in models of bacterial nitrogen signaling,” as a Research Article in PLOS Computational Biology.

The study asks which experimental decisions remain possible when different enzyme mechanisms predict the same target response. Using the GlnD–PII–GlnE–GS cascade, it links explicit mechanistic equivalence to a concentration decision and tests a complementary measurement from raw titration points through statistical discrimination.

First, finite binding–catalysis transformations preserve complete stationary target-state distributions or downstream responses. Second, a concurrent PII modification count sharpens a conservation-based bound on differences between total GS observations. In a source-anchored prospective example, the general bound is 0.370 while the observation-conditioned bound is 0.041, sufficient for a prescribed 0.05 tolerance after a fivefold reduction in GlnE. The result does not require independent PII sites or a known downstream catalytic gain. Its practical value is the concentration choice it can certify when a general bound cannot.

Third, a binding-assay benchmark includes finite ligand depletion, shared baselines, enzyme-total errors, systematic dose bias and estimated variance. Independent concentration standards correct the tested bias, but a hidden affinity mixture still produces misleading single-class mechanistic attribution despite small residuals. A separate GlnD constraint profile shows that improved titration agreement can reduce the occupancy contrast targeted by an assay. These results make biochemical constraints and measurement controls part of the decision itself.

The work is methodological and theoretical; no new experimental measurements were generated. Source assay conditions and a conflicting supplementary unit are documented explicitly. The concentration gate and coefficient envelope require confirmation in the proposed assay. Neither finite-model checks nor synthetic discrimination validate the complete physiological cascade. A paired-paralog application is presented as prospective because matched calibration transfer and direct occupancy validation are not yet available.

The manuscript contains five main figures, two appendices and six supporting figures. Data, executable analyses, numerical checks and manuscript sources accompany the synchronized version. Established stationary elimination and identifiability principles are acknowledged as foundations; the contribution is their connection to a biochemical interpretation guarantee and an observation-level design benchmark.

I believe this explicit connection between mechanistic ambiguity, measurable constraints and experimental decisions will interest readers studying signaling and quantitative experiment design.

Thank you for considering the manuscript.

Sincerely,

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